BrineWatch

Seeing what desalination leaves on the seabed

An underwater robotic system for outfall inspection and brine-plume mapping

An underwater robot that locates desalination outfalls, maps the invisible brine plume around them and tells operators where environmental sampling matters most, turning a routine inspection dive into actionable environmental evidence.

Applicant & team lead	**Andrea Bedei**, PhD student, University of Bologna
Academic supervisors	Roberto Girau · Giovanni Pau · University of Bologna
Project collaborators	Lorenzo Bacchiani · Andrea Melis
Project phase	**Simulation Prototype or Model**
Submission	Prototypes for Humanity 2026

Cover image: the simulated outfall diffuser and inspection scene used to develop and test the full BrineWatch mission.

1 The problem: fresh water's invisible waste stream

More than 300 million people drink water that comes from the sea. Desalination keeps entire regions alive. It is growing fast in the Gulf and increasingly across the Mediterranean. But every litre of fresh water leaves behind roughly 1.5 litres of **brine**: seawater concentrated to nearly twice its natural salinity, often warmer, discharged back into the sea through submarine outfalls, amounting to about **142 million m^3 every day** worldwide, more than half of it in the Gulf region alone [1].

The physics is what makes this problem invisible. Brine is *denser* than seawater: after leaving the diffuser it sinks, collapses onto the seabed and creeps along the bottom as a slow salty layer [2]. That is exactly where seagrass meadows, shellfish and the base of coastal food chains live. This is exactly where nobody is looking. **Satellites see only the surface. Fixed buoys measure one point, usually at the wrong depth. Survey vessels visit a few times a year.** Between visits, the plume moves with tides and currents, unobserved.

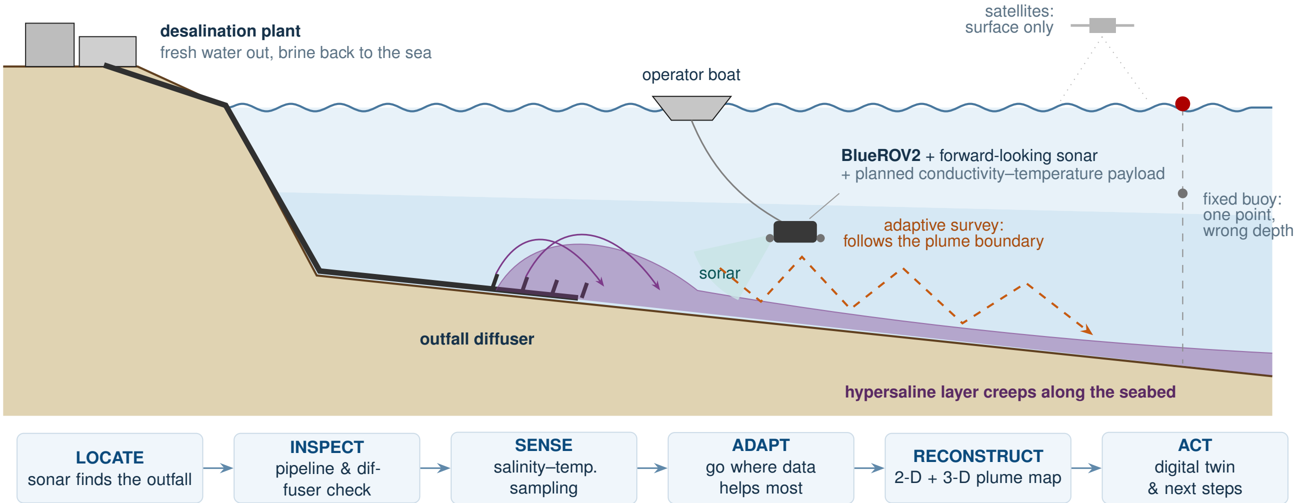


Figure 1: The BrineWatch mission at a glance. Dense brine sinks and spreads along the seabed, where satellites, buoys and quarterly boat surveys cannot follow it. BrineWatch sends a small robot along the plume itself: it finds the outfall with sonar, inspects the structure, samples salinity where it is most informative and returns a mapped, uncertainty-aware picture of the discharge zone.

Who feels this problem Desalination plant operators · environmental regulators · marine survey companies · coastal authorities · research laboratories

2 Why today's monitoring falls short

Discharge permits define a **mixing zone**: a distance from the diffuser beyond which salinity must return close to ambient. Verifying it requires spatial information: where the plume actually goes. Yet routine monitoring still relies on tools that were never designed to answer a spatial question:

- **Fixed stations**: accurate, but only at a handful of points that the plume can simply miss;
- **Manual probe casts**: a technician lowers an instrument from a boat, station by station;
- **Vessel campaigns**: thorough but expensive, so they happen every few months;
- **Static reports**: results arrive weeks later, describing a sea state that has already changed.

The result: operators and regulators work with a few dozen correct numbers that may still miss the plume between them.

TODAY'S WORKFLOW

- sparse point measurements
- fixed locations, chosen in advance
- vessel campaign every 3–6 months
- results reported weeks later
- little or no usable site history

THE BRINEWATCH WORKFLOW

- mobile measurements along the plume
- survey adapts to what it finds
- inspection + environment in one mission
- map with explicit uncertainty, same day
- repeatable digital site history

What changes in practice

Consider the routine question: *"is the discharge behaving as permitted?"* Today the answer depends on whether a handful of fixed points happened to intersect the plume. With BrineWatch, one robot mission produces a picture: where the outfall is, the condition of the diffuser, a salinity map around it with a clear boundary estimate, the parts of the site that remain uncertain and a concrete recommendation of where to send the certified sampling team next. The mission is repeatable, so each visit adds a page to the same site history rather than starting from zero.

Figure 2: Approaching the outfall. A frame from the project's simulated inspection mission: the pipeline and diffuser risers emerge from the haze as the vehicle approaches. This is the moment monitoring stops being blind.

3 The BrineWatch system

BrineWatch is built deliberately from accessible parts: a proven open-source inspection robot, one commercial imaging sonar, one scientific sensor and software that turns them into an autonomous environmental instrument.

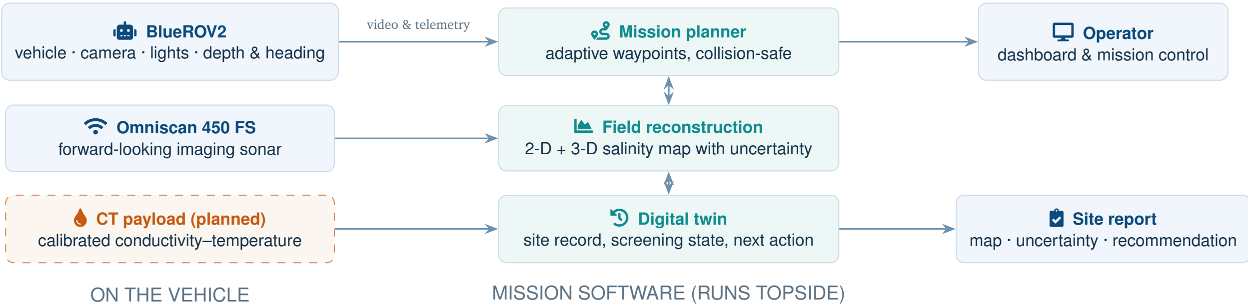


Figure 3: System overview. Solid blue panels: hardware available today. Dashed panel: the calibrated CT sensor, the next integration step. Teal panels: mission software already implemented and exercised in simulation.

ALREADY IN PLACE

- BlueROV2 owned by the team
- full mission software: planning, sonar localisation, mapping, digital twin
- high-fidelity simulation environment (custom HoloOcean fork, Unreal Engine)
- complete simulated missions with video evidence

NEXT INTEGRATION STEPS

- validation in controlled water, then at sea

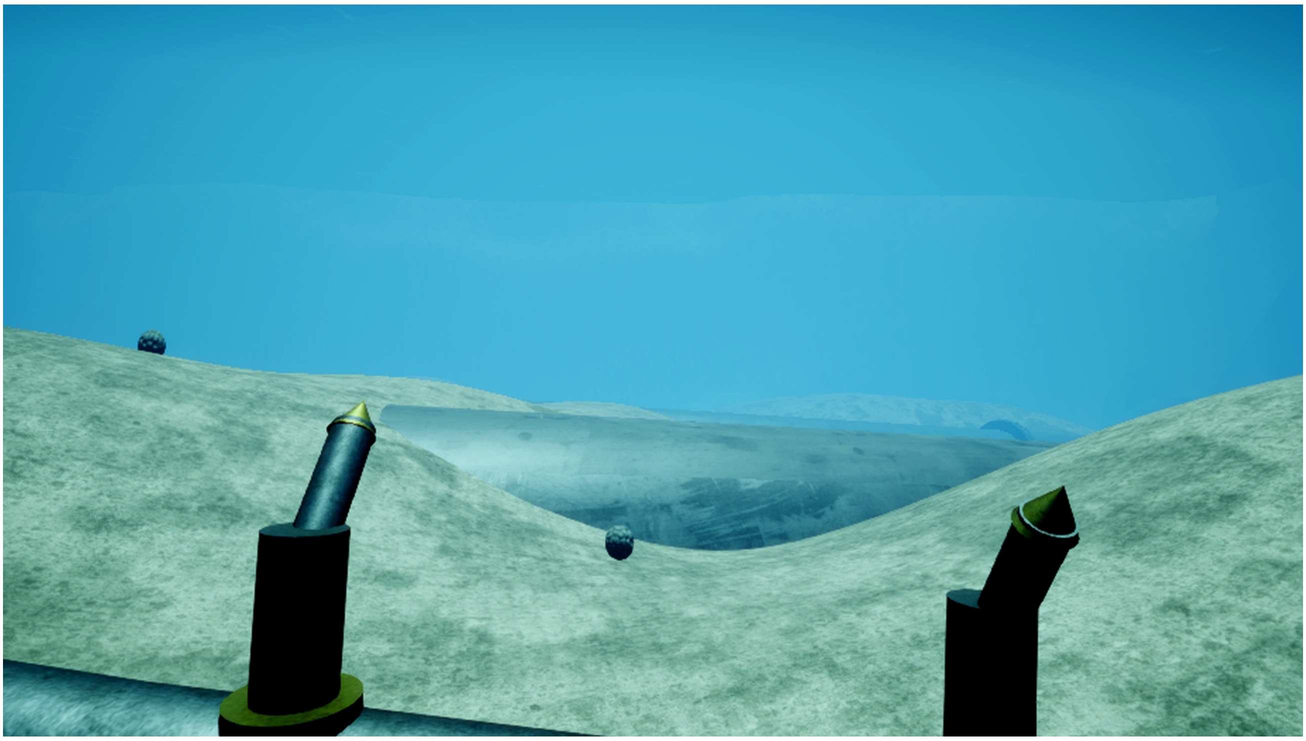


Figure 4: Inside the simulation. The diffuser risers seen from the vehicle during a simulated inspection pass.

4 How a mission works

Every BrineWatch mission follows the same six movements: simple enough to brief a crew in a morning, rich enough to produce evidence a regulator can use.

LOCATE

The forward-looking sonar finds the outfall on the seabed and anchors the whole survey to it. No GPS exists underwater.

INSPECT

The vehicle follows the pipeline to the diffuser, checking risers and nozzles with camera and sonar, at a safe standoff.

SENSE

The CT payload records salinity, temperature and depth continuously along the route.

ADAPT

After each leg, the planner asks: *where would the next reading teach us most?* It then routes the vehicle there.

RECONSTRUCT

All samples fuse into 2-D and 3-D salinity maps with an explicit confidence level for every location.

ACT

The digital twin stores the mission, raises a clear screening state and recommends where certified sampling should go next.

Figure 5: One continuous simulated mission. Clockwise from top left: descent to the site; close pass over the diffuser nozzles; the sonar's view, where the bright arc is the outfall structure as the localisation algorithm sees it; following the pipeline during inspection. All frames come from the same in-engine mission that produced the evidence in Section 6.

5 What the digital twin actually is

Digital twin can sound like a buzzword, so here is the plain meaning:

The BrineWatch digital twin is the evolving operational record of one outfall site.

It is not a decorative 3-D graphic. It is a structured, queryable memory: every mission adds its sonar fix, its samples, its route, its reconstructed maps and its conclusions to the same record. The site's story accumulates instead of evaporating into filed-away PDFs.

Figure 6: The twin in one picture. Missions feed the model on the left; operators and regulators read answers on the right.

From mission data to the next action

BrineWatch does not issue a legal pass/fail verdict. Instead, after each mission it combines the reconstructed salinity map with its uncertainty and assigns one of three operational screening states. **CLEAR** means the plume boundary is sufficiently resolved and no elevated salinity is indicated beyond the mixing zone, so routine monitoring can continue. **REVIEW** means the evidence remains insufficient; the uncertainty map identifies where more observations are needed. **POSSIBLE EXCEEDANCE** means elevated salinity is reconstructed beyond the mixing-zone threshold with sufficient confidence, prompting targeted certified reference sampling at the highest-risk location. These states support, rather than replace, regulatory assessment by turning mission data into a clear and traceable next action.

The dashboard in Figure 7 presents this workflow to the operator. Its top row summarises the current screening state and key quality indicators. The map shows the reconstructed salinity field, estimated plume boundary and robot trajectory; the central panel highlights uncertainty; and the right-hand panel converts the evidence into an operational recommendation. In the example, **POSSIBLE EXCEEDANCE** triggers targeted certified sampling at the highest-risk plume-boundary location. The timeline preserves previous mission outcomes so operators can track change over time and each survey contributes to the same digital record.

Figure 7: BrineWatch operator dashboard. The dashboard links the latest plume map, robot trajectory, uncertainty, screening state, mission history and recommended next action in a single operational view.

6 What has been built and what it shows

BrineWatch today is a **simulation-led prototype**. The complete system, including vehicle control, sonar, planning, mapping and the digital twin, runs end-to-end in a high-fidelity underwater simulator (a custom fork of HoloOcean [4], built on Unreal Engine).

6.1 A full robotic mission inside the engine

One continuous mission in the custom simulator demonstrated the complete robotic workflow using native vehicle dynamics and native simulated sonar. Figure 8 isolates the navigation sequence to make the evidence easy to read. The grey line marks the outfall structure and the six circles mark the diffuser risers. The grey cross is the approximate chart prior and the orange star is the sonar fix obtained by the vehicle. From that anchor, the pale route records the baseline legs and the dark-blue route shows the adaptive survey around the structure. This is the same mission summarised below:

394.8 m

travelled in-engine, within a 400 m budget

564

salinity readings collected along the route

0

collisions, with 4 safe detours around the structure

2.35 m

sonar localisation error of the diffuser

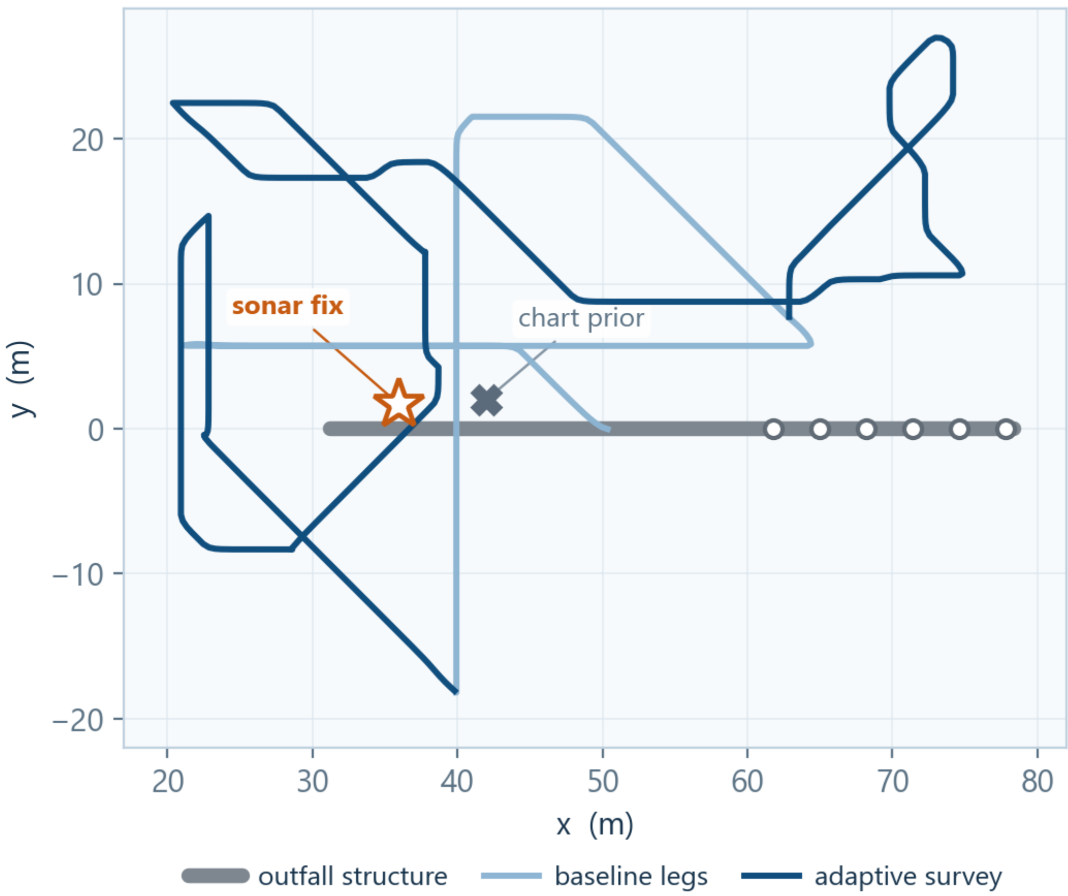


Figure 8: **In-engine navigation sequence.** Sonar localisation, baseline legs and adaptive survey around the outfall.

6.2 A controlled demonstration of the science

Figure 9 compares a known analytic plume on the left with the field reconstructed by BrineWatch from 591 samples in the same controlled mission on the right. The white line is the robot trajectory, the orange contour is the boundary estimated by the system and the dashed dark contour is the known threshold boundary. The reconstruction recovers the plume's elongated footprint and keeps the less-observed outer areas visibly less regular instead of presenting them as equally certain. In this scenario the estimated boundary extends beyond the screening contour, so the system correctly returns **POSSIBLE EXCEEDANCE**. This demonstrates the complete mapping and screening chain under controlled conditions.

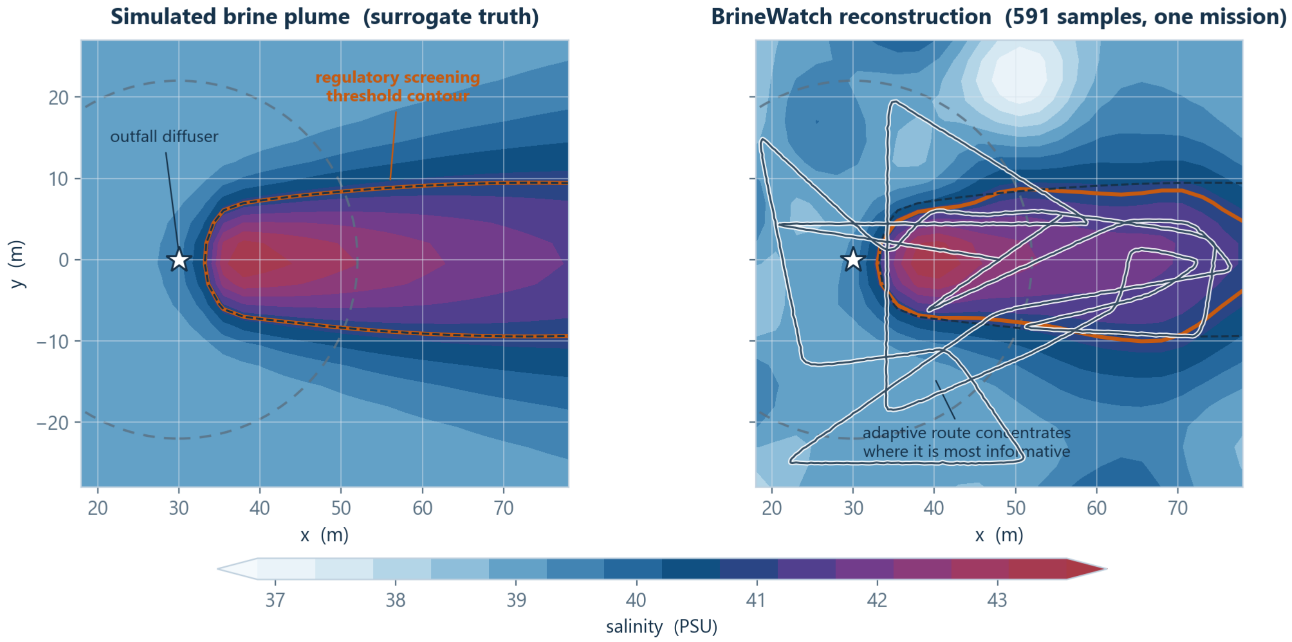


Figure 9: Controlled plume reconstruction from 591 samples.

6.3 The plume as a volume

Because brine hugs the seabed, a single-depth map can miss its vertical structure. Figure 10 extends the same controlled scenario into three dimensions using 912 samples collected across four altitude bands from 0.8 to 4.5 m above the bed. The grey bar marks the diffuser, the thin lines show the four survey paths and the coloured cells are the reconstructed water volume above the screening threshold. The densest region remains close to the seabed and narrows with height, while sparse areas away from the sampled bands remain visibly uncertain. The model reconstructs 2,051 m^3 of above-threshold water against a simulated truth of 2,448 m^3, with a volume overlap of 0.81 / 1 and a typical salinity error of 0.48 PSU.

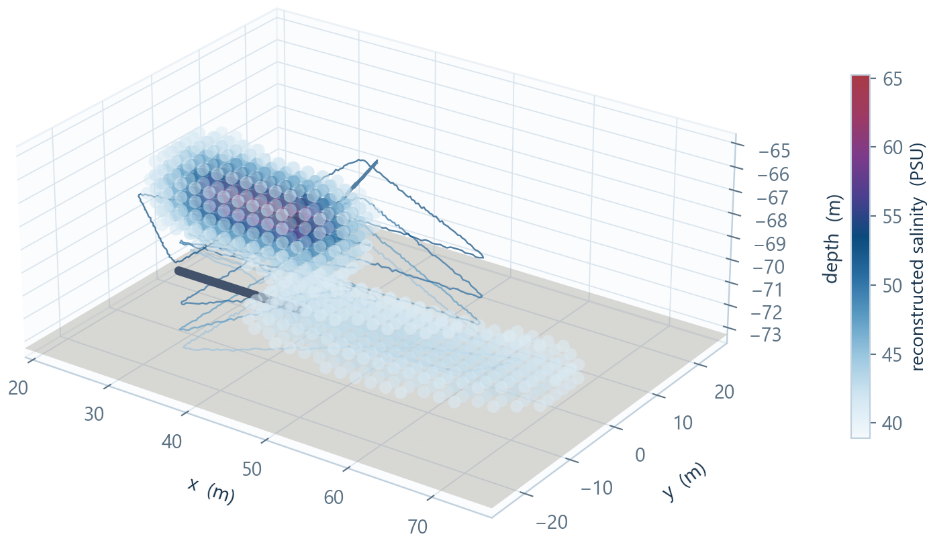


Figure 10: Three-dimensional plume reconstruction from four survey altitudes.

7 Why the adaptive route matters

Is the adaptive planner actually worth it? We tested exactly this with an equal-evidence benchmark: **same simulated plume, same 48 readings, same 300 m travel budget and eight repetitions per method**. Only the sampling strategy differs.

Figure 11 shows the outcome. Sparse fixed stations produced 17% useful readings and no conclusive missions. The regular survey pattern increased this to 23% useful readings and four conclusive missions out of eight. The adaptive route concentrated 70% of its readings in the informative region, reduced the unresolved plume area to 28% and produced eight conclusive missions out of eight. The paths are illustrative, while the percentages and mission counts are computed benchmark results. No method produced a false **CLEAR**.

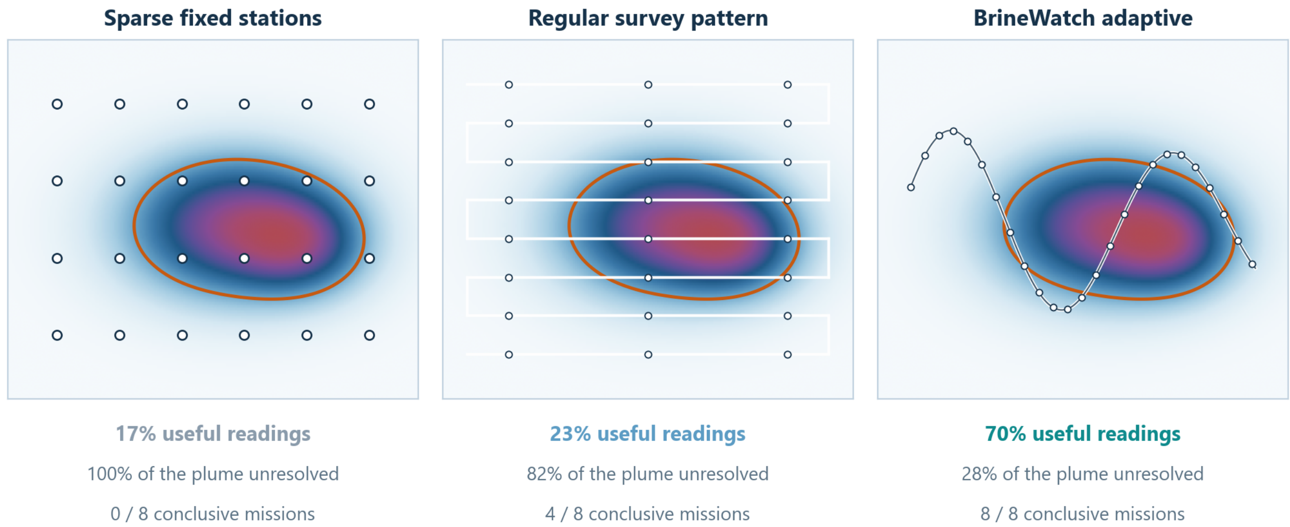


Figure 11: Equal-evidence comparison of three sampling strategies.

8 Technical feasibility

BrineWatch was designed backwards from feasibility: standard vehicle, commercial sonar, one low-risk scientific sensor and software that is already written and exercised. What separates today's prototype from a sea-going instrument is integration and validation.

EXISTS AND WORKS TODAY

- BlueROV2 (owned); Omniscan 450 FS selected, commercially available
- custom simulation environment with native imaging sonar
- sonar localisation with a 2.35 m error
- collision-safe adaptive mission planning
- 2-D + 3-D reconstruction with uncertainty
- digital twin with three-state screening
- mission video and full evaluation package

NEXT STEPS TOWARDS FIELD DEPLOYMENT

- calibrated CT sensor integration
- time synchronisation and calibration
- controlled-water and harbour validation
- permits, site access and nearshore pilot partner

Readiness, honestly stated

The software stack is mission-complete in simulation. The hardware platform is proven commercial equipment. A calibrated CT sensor is standard oceanographic equipment, but *integrating and validating* it is exactly the work the next phase funds. Simulation results use an analytic plume surrogate, not CFD or field truth: the entire purpose of the roadmap is to replace that surrogate with real water, step by step.

Main risks and how we contain them

Risk	Likelihood	Impact	Mitigation
CT calibration drift	M	H	Bench and controlled-water calibration; cross-checks with a reference probe during each campaign
Sonar localisation in clutter	M	M	Multi-radius consensus already implemented; operator-assisted anchoring available as a fallback
Water turbidity	M	L	The mission relies primarily on sonar; the camera is secondary
Currents and station keeping	M	M	Heavier vehicle configuration, conservative survey speeds and predefined abort thresholds
Navigation near the structure	L	H	Collision-safe planner validated in simulation, with zero collisions and four safe detours; conservative standoff distance
Regulatory approval and site access	M	H	Early engagement with utilities and regulators; harbour trials can proceed without access to an operating outfall
Field reference sampling	L	M	The pilot budget includes independent certified reference measurements

Scope statement. BrineWatch is a screening and targeting instrument on a path to controlled-water and nearshore validation. It is not a certified compliance system. Simulation results are not field measurements; they demonstrate the instrument, not the ocean.

9 Cost, management and roadmap

Reusable system, low-cost repeat missions

BrineWatch separates the one-time cost of completing the robotic system from the much lower cost of repeating a mission. It is completed once and reused: subsequent missions require no dedicated survey vessel and produce a precise spatial map rather than a limited set of isolated measurements.

After the initial investment, recurring costs are limited to local transport, consumables, equipment wear and the internal operational team. When needed, a small local boat can be rented for a short deployment. BrineWatch does not require the purchase or permanent operation of a fully equipped survey vessel.

Cost model	Indicative cost	Main result
One-time system completion	**USD 5k–9k once**	Reusable ROV, sonar, CT payload and digital-twin system
Repeated BrineWatch mission	**USD 150–500 from shore; USD 1k–2.5k with a small chartered boat**	Inspection, adaptive measurements, plume map, uncertainty and site-history update
Fully equipped research vessel	**USD 20.5k–24.5k per day, vessel only**	Conventional ship-based survey

ONE-TIME SYSTEM COMPLETION
USD 5k–9k

Omniscan 450 FS, calibrated CT payload, mounting and interfaces, integration, calibration and essential spares. Once completed, the same system can be reused across repeated missions.

REPEATED BRINEWATCH MISSION
USD 150–500 from shore
USD 1k–2.5k with a small chartered boat

CONVENTIONAL VESSEL CAMPAIGN
USD 20.5k–24.5k per day
for the research vessel alone

Published 2026 rates for fully equipped research vessels are approximately USD 20.5k–24.5k per day, before additional scientific personnel, laboratory analysis, permits and reporting [7].

Lower cost without sacrificing spatial evidence

BrineWatch's advantage is not only economic. One mission combines outfall localisation, infrastructure inspection, adaptive environmental measurements, 2-D and 3-D plume reconstruction, uncertainty estimation and a digital-twin update.

In the controlled simulation benchmark, BrineWatch achieved a plume-boundary score of **0.95/1**, a 2-D overlap of **0.90/1** and a salinity reconstruction error of **0.34 PSU**. The 3-D reconstruction achieved an overlap of **0.81/1** with an error of **0.48 PSU**.

These results demonstrate high reconstruction accuracy in controlled simulation. Field accuracy will be independently validated during controlled-water and nearshore trials.

Development roadmap with decision gates

- **P1: CT payload integration** (0–3 mo): sensor, mounting, power, time sync and bench calibration. *Main cost: sensor + integration.*
- **P2: Controlled water** (3–6 mo): known salinity gradients, independent truth samples, reconstruction-error validation and repeatability. *Main cost: facility time.*
- **P3: Harbour / sheltered water** (6–9 mo): navigation robustness, sonar localisation on real structures and surrogate plume-release experiments. *Main cost: operations.*
- **P4: Supervised nearshore pilot** (9–15 mo): a real outfall inspection with certified reference samples and operator feedback. *Main cost: vessel, permits and reference lab.*
- **P5: Repeat monitoring** (15–24 mo): scheduled campaigns, trends in the twin and service definition. *Main cost: campaign operations.*

10 Applications, impact and what support unlocks

PRIMARY APPLICATIONS

- desalination outfall inspection
- brine-plume screening around diffusers
- targeting of certified environmental sampling
- repeated digital site history per outfall

THE SAME INSTRUMENT ALSO SERVES

- wastewater outfalls and industrial discharge points
- desalination intakes and their protection zones
- harbour and coastal infrastructure inspection
- research monitoring of benthic habitats

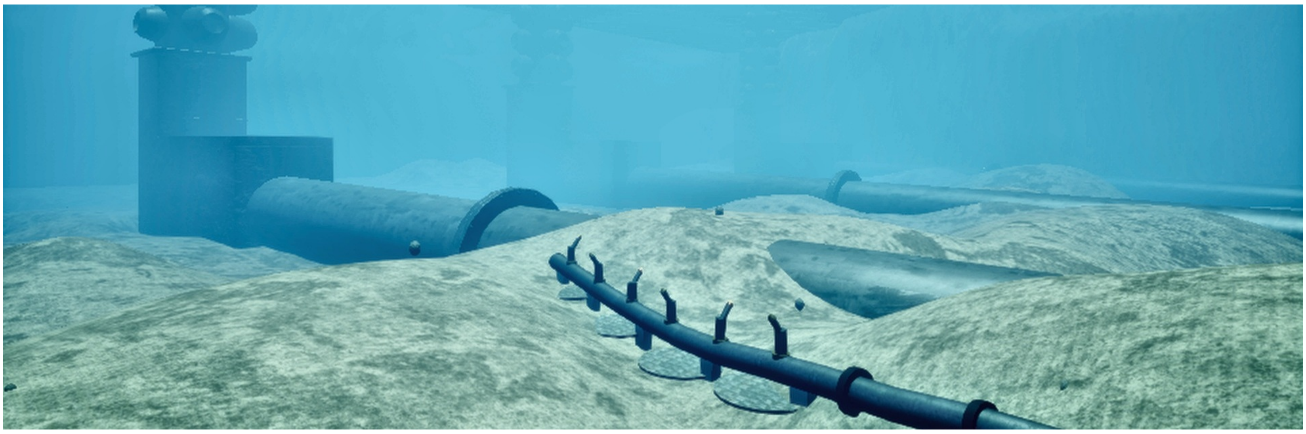
Impact, in four dimensions

Environmental	seabed conditions become visible; certified sampling lands where it matters; anomalies surface earlier, between formal campaigns.
Operational	inspection and environmental screening merge into one mission; site understanding same-day instead of next-quarter; a procedure any trained ROV crew can repeat.
Economic	reuses robotic hardware many operators already own; fewer blind sampling stations; growing value of an accumulated site history.
Scientific	a working bridge between CFD predictions, robotic measurement and field validation, with a reusable architecture for underwater environmental monitoring.

What support would enable

With the resources of the next validation gate (Section 8), the team will: integrate and calibrate the **CT payload**; validate reconstruction accuracy in **controlled water** against independent truth; run **harbour trials**; and prepare a **supervised nearshore pilot** at an outfall with certified reference sampling.

BrineWatch is ready to move from simulation-led evidence to controlled-water and nearshore validation.



Our outfall site in the simulation.

Andrea Bedei · University of Bologna · supervisors Roberto Girau & Giovanni Pau · Prototypes for Humanity 2026

[1] E. Jones *et al.*, “The state of desalination and brine production: a global outlook,” *Sci. Total Environ.* 657, 2019.

[2] P. J. W. Roberts, “Near-field flow dynamics of concentrate discharges and diffuser design,” in *Intakes and Outfalls for Seawater RO Desalination*, Springer, 2015.

[3] G. Hitz *et al.*, “Adaptive continuous-space informative path planning for online environmental monitoring,” *J. Field Robotics* 34(8), 2017.

[4] E. Potokar *et al.*, “HoloOcean: an underwater robotics simulator,” *IEEE ICRA*, 2022; “A preview of HoloOcean 2.0,” arXiv:2510.06160, 2025.

[5] Blue Robotics: BlueROV2 product page, price checked 22 July 2026.

[6] Cerulean Sonar: Omniscan 450 FS product page, price checked 22 July 2026.

[7] Marine Institute: 2026 vessel rates of EUR 18,000–21,500 per day; converted at the ECB reference rate of USD 1.1426 per EUR, 20 July 2026.

[8] BrineWatch project video: `https://youtu.be/bsZbojE48Yk`.

All performance figures in this document are simulation results, reproduced deterministically from the project repository; the technical evidence ledger with full metrics and limitations accompanies this proposal.